higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE

# INDUSTRIAL ELECTRONICS N5

(8080175)

**4 August 2023 (X-paper)**
**09:00–12:00**

**Drawing instruments and nonprogrammable calculators may be used.**

**This question paper consists of 6 pages and a formula sheet of 6 pages.**

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# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

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**INSTRUCTIONS AND INFORMATION**

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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## QUESTION 1: ALTERNATING CURRENT THEORY

1.1 Draw a neat, labelled diagram of a coupling circuit. (3)

1.2 Indicate the input and output waveforms of the circuit drawn in QUESTION 1.1. (2)

1.3 What is the main purpose of a coupling circuit? (2)

1.4 A series RLC network connected across an AC supply voltage has the following component specifications:

R = 100 Ω

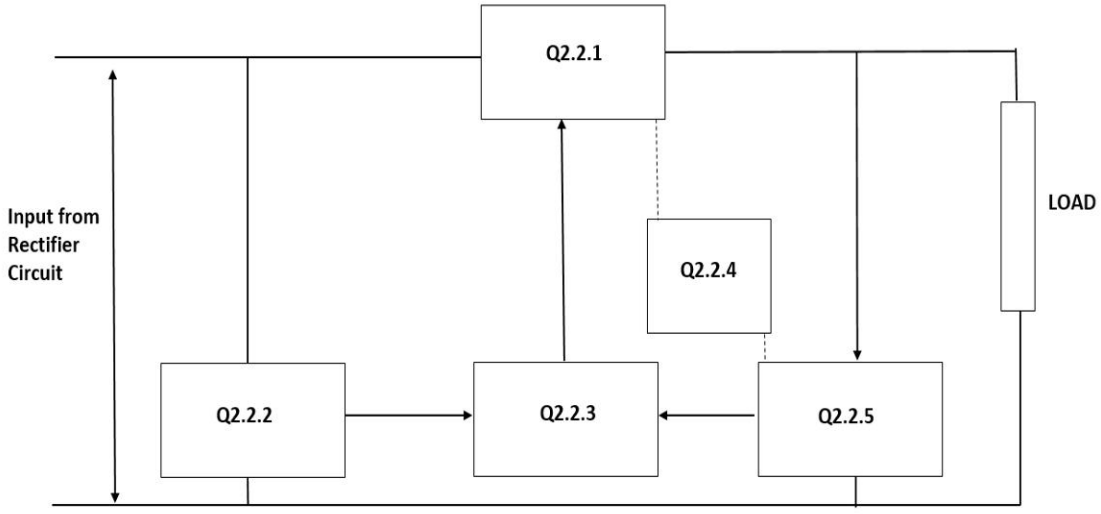
L = 0,2 H

C = 47 µF

Determine the lower- and upper-half power frequencies. (8)
**[15]**

## QUESTION 2: POWER SUPPLY

2.1 Supply the labels for the blocks in the figure below. Write only the answer next to the question number (2.2.1–2.2.5) in the ANSWER BOOK.




(5)

2.2 Name the type of regulator show in the figure above. (1)

2.3 Briefly explain the following terms with reference to rectifier circuits:

2.3.1 No-load voltage

2.3.2 Voltage regulation

2.3.3 Filter

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2.3.4 Voltage doubler

(4 × 2) (8)

2.4 State TWO practical applications of a full-wave rectifier. (2)

**[16]**

## QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a common emitter amplifier are known:

+Vcc = 12 V  $I_C$ = 10 mA  Vce = 6 V
$\beta$ = 250  Re = 120 Ω  Vbe = 0,6 V

Calculate the values of the following components:

3.1.1 RC (collector resistor) (4)

3.1.2 $R_{b1}$ and $R_{b2}$ (6)

3.2 Calculate the input impedance $Z_i$ and the output impedance $Z_O$ of the circuit in QUESTION 3.1 by means of the approximate method if:

$h_{ie}$ = 1,2 kΩ  $h_{re} = 2 \times 10^{-4}$

$h_{fe}$ = 100  $h_{oe}$ = 20μA/V ($R_S$ = 0) (4)

3.3 What effect does a temperature rise have on the following?

3.3.1 The collector current $I_{CO}$

3.3.2 The base-emitter voltage $V_{be}$

(2 × 2) (4)

**[18]**

## QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 With reference to operational amplifiers, give a short description of *drift*. (3)

4.2 Draw a neat, labelled circuit diagram of a practical integrator. (4)

4.3 What is the purpose of a voltage adder? (1)

4.4 List THREE important properties of an operational amplifier. (3)

**[11]**

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## QUESTION 5: INTEGRATED CIRCUITS

5.1 State TWO types of protection networks used to protect the gate oxide of the CMOS against high levels of electrostatic discharges. (2)

5.2 Draw a full waveform of the voltage pulses that exceed $\pm 15$ V to illustrate the clamping provided by the Zener diode. (3)

5.3 Name TWO procedures that must be followed when handling CMOS integrated circuits. (2)

**[7]**

## QUESTION 6: ELECTRONIC PHASE CONTROL

6.1 What is the purpose of the trigger circuit in a phase trigger circuit?

6.2 Show with the aid of a sketch of a waveform the output of a full-wave AC control with one SCR of a bridge rectifier.

(2 × 2) **[4]**

## QUESTION 7: TEST EQUIPMENT

Draw a neat, labelled block diagram of a frequency counter. **[5]**

## QUESTION 8: OSCILLATORS

8.1 State the use of Rozner modification with respect to the astable multivibrator. (2)

8.2 Show with the aid of a well-labelled waveform how the Rozner modification in QUESTION 8.1 is achieved. (2)

8.3 An astable multivibrator has the following parameters:

$R_A = 20\ k\Omega$
$R_B = 5\ k\Omega$
$C = 47\ \mu F$

Calculate the oscillator frequency of the multivibrator. (3)

8.4 Draw a well-labelled equivalent circuit diagram of a quartz crystal. (2)

8.5 State TWO criteria that an oscillator circuit must fulfil irrespective of the type of waveform it produces. (2)

**[11]**

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## QUESTION 9: TRANSDUCERS

9.1 Choose a description from COLUMN B that matches the transducer type in COLUMN A. Write only the letter (A–H) next to the question number (9.1.1–9.1.7) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 9.1.1 | Resistance strain gauge | A | sound pressure varies the capacitance between a fixed plate and a movable diaphragm |
| 9.1.2 | Resistance thermometer | | |
| 9.1.3 | Dielectric gauge | B | secondary electron emission will occur due to incident light on a photo-emissive surface |
| 9.1.4 | Capacitor microphone | | |
| 9.1.5 | Photomultiplier | C | resistance wire with high positive temperature coefficient changes with a variance in temperature |
| 9.1.6 | Photovoltaic cells | | |
| 9.1.7 | Magnetostriction gauge | D | an emf will be generated when incident light strikes the surface of a semiconductor junction |
| | | E | magnetic properties are varied |
| | | F | variation in capacitance by variation of the dielectric medium |
| | | G | resistance of the wire changes when subjected to elongation or compression due to external stress being applied |
| | | H | resistance of cell changes with a variance in incident light |

(7 × 1) (7)

9.2 What is the difference between an *active transducer* and a *passive transducer*? (2)

9.3 A thermistor has values of 12 kΩ at 10 °C and 2 kΩ at 30 °C.

Calculate the value of the constant B. (4)

**[13]**

**TOTAL: 100**

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## FORMULA SHEET

$$I = \frac{V}{R}$$

$$P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC$$

$$T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt}$$

$$V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL$$

$$X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L$$

$$Z = R - jX_C$$

$$Z = R + j(X_L - X_C)$$

$$I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR$$

$$V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C)$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1$$

$$\frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2}$$

$$I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L}$$

$$\frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L$$

$$I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C}$$

$$\frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C$$

$$I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C)$$

$$I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \underline{\tan^{-1}\frac{b}{a}} = r\,\underline{/\theta}$$

$$r\underline{/\theta} = r(\cos\theta + j\sin\theta)$$

$$Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}V_m = 0{,}707\,V_m$$

$$V_{dc} = \frac{2}{\pi}V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

$$V_{dc} = \frac{1}{\pi}V_m = 0{,}318\,V_m$$

$$PIV = V_m$$

$$PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m$$

$$V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L}.V_{dc}$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}}.V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R}.V_{r(rms)}$$

$$r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$$TV'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

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$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L} \qquad Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{h_{fe}{}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe} \left( \frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right) \qquad A_i = \frac{I_0}{I_1}$$

*For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.*

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log \frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left( \frac{R_f}{R_1} \right) . V_i \qquad V_0 = \left( \frac{R_f}{R_1} + 1 \right) . V_i$$

$$V_0 = -\left( \frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right) \qquad V_0 = -\left( \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\ R_f$$

$$V_0\ (t) = -\frac{1}{RC} \int V_i(t) \qquad V_0\ (t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \qquad A_v = -\frac{R_s}{R_1}$$

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$$R_2 = \frac{R_1 R_s}{R_1 + R_s}$$

$$f_c = \frac{1}{2\pi R_s C}$$

$$V_0(t) = -RC\frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C \frac{d}{dt}.v_i \sin \omega t$$

$$t = R_f C$$

$$V_0 = A(V_r - V_i)$$

$$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$$

$$V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$$

$$L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1,5}{RC}$$

$$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0,7\ R_2 C_1$$

$$t_2 = 0,7\ R_1 C_2$$

$$f_0 = \frac{1}{1,4RC}$$

$$V_i = I_{c2} R_e + V_{be(ON)}$$

$$t = 1,1\ RC$$

$$f_0 = \frac{1,443}{(R_A + 2R_B)C}$$

$$t_{low} = 0,693\ (R_B)C$$

$$t_{high} = 0,693\ (R_A + R_B)C$$

$$t_T = t_{low} + t_{high}$$

$$K = \frac{\Delta R / R}{\Delta l / l}$$

$$\sigma = \Delta l / l$$

$$R = \rho \frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E}$$

$$Resolution = \frac{1}{amount\ of\ turns}$$

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$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$R_t = Ae^{B/T}$

$T = 273 + °C$

$V_A = \frac{R_2}{R_1 + R_2} . V_T$

$V_B = \frac{R_t}{R_t + R_3} . V_T$

$V_{AB} = V_A - V_B$

$A_v = \frac{V_0}{V_i}$

$V_{Hall} = kIH$

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